NuMicro® Family

Arm® Cortex®-M4-based Microcontroller

M471 CMSIS BSP
Revision History

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Revision 3.04.000 (Released 2026-10-16)

1. Fix FMC_IAP VSCode Sample coode build error
2. Fix Doxygen warnings, comments, and typos
3. Merge "Modify some drivers to fix the MISRA violation."
4. Modify some drivers to fix the MISRA violation.
5. Fix FMC_IAP VSCode Sample code Build Error
6. Fix ISP VSCode Sample code Build Error
7. Update retarget.c to fix VSCode GCC warning message
8. Merge "Modify clk.h/timer.h/timer.c to fix the TIMER related MISRA violation"
9. Modify clk.h/timer.h/timer.c to fix the TIMER related MISRA violation
10. Modify gpio.c to fix the GPIO related MISRA violation
11. Merge "[SBOM] Correct component identity and dependency evidence"
12. [SBOM] Correct component identity and dependency evidence
13. Merge "Fix MISRA-C violations for I2C, PDMA, UART and PRNG driver."
14. Fix MISRA-C violations for I2C, PDMA, UART and PRNG driver.
15. [SBOM] Synchronize V3.03.000-21 release artifacts
16. [SBOM] Add M471 current SBOM evidence
17. Update retarget.c, system.c and startup.c for misra-cppcheck
18. Add FreeRTOS sample code
19. Modify dac/spi/bpwm/epwm/timer_pwm driver to fix some MISRA violations.
20. Correct RTC reference definition for frequency compensation
21. Fix FMC/DFMC/RTC/WDT MISRA C violations
22. Update VSCode Project setting
23. Fix include filename case in CIR sample code for NuEclipe
24. Update CIR driver for MISRA-C/Cppcheck compliance and improve CIR sample code readability.
25. Fix update DataFlash fail issue
26. Fix FMC multi-word programming 1. Remove redundant code 2. Correct return value handling on timeout
27. Fix FMC Multi-Word Program 1. Fix BSP driver missing status check for the final 4-word write. 2. Correct the retransmission address calculation for timeout recovery. 3. Update the maximum multi-word program size from 512 to 256 bytes. (by DTS) 4. The scatter file was modified due to increased SRAM usage.
28. Modify PDMA EnableInt function to support two more mask at the same time
29. Fixed the Keil project so that NU_LINK debugger is selected by default when opening the .uvprojtx file.
30. 1. Revised i2c driver - Extended multiple bytes read api to support read single byte - Merged single byte / multiple byte api implementation. 2. Fixed ISP_I2C/ISP_SPI - Disabled IRQ until ParseCmd() is completed to prevent ISP from triggering an interrupt and fetching data prematurely. 3. Fix typo.
31. Update cproject file wit NuMicroM4_DFP instead of NuMicro_DFP
32. Set cmake tool to version 3.31.5 to fix long path name issue
33. Add a check for the STOP bit when I2C transmission is complete.
34. Add comment and control flow for issue #10338
35. Addd comment and control flow for issue #10338
36. Update settings.json for project name

Revision 3.03.000 (Released 2024-12-17)

37. Upgraded CMSIS from V5.1.1 to V6.1.0.
38. Supported VSCode for AC6 and GCC.

39. Updated setting order of scatter-gather mode for PDMA driver.
40. Minor changes for sample code.
41. Minor bug fix.

Revision 3.02.000 (Released 2024-10-30)

42. Added EADC_BandGapCalculateAVDD sample code.
43. Added FMC IRQHandler definition for GCC and IAR startup codes.
44. Added .uvprojx project file for Keil5.
45. Updated I^2C driver to support 10 bits slave address mode.
46. Updated SPI driver to support slave 3-wire mode.
47. Removed .uvproj project file for Keil4.
48. Minor changes for sample code.
49. Minor bug fix.

Revision 3.01.000 (Released 2023-05-18)

1. Added BPWM_GET_PRESCALER(), GPIO_SET_DEBOUNCE_ICLKON() and GPIO_CLR_DEBOUNCE_ICLKON() macro definitions.
2. Added EADC_SetVRef() API.
3. Added EADC_Accumulate, EADC_Average and EADC_Vref sample code.
4. Added DFMC_NonBlocking sample code.
5. Added ACMP Calibration API.
6. Added CPSCBUF[] and IFACNT[] registers definition in EPWM_T structure.
7. Added V_{REF} register definition in DAC_T structure.
8. Added V_{REF} register definition in EADC_T structure.
9. Removed INTERNAL register definition in SPI_T structure.
10. Removed ACMP_CTL_HYSTERESIS_50MV, ACMP_CTL_HYSTERESIS_30MV and ACMP_CTL_HYSTERESIS_10MV definitions.
11. Set DFMC access cycle according to HCLK frequency.
12. Enable I^2C pin Schmitt trigger in I^2C sample code.
13. Enable GPIO ICLKON bit for power consumption in SystemInit().
14. Minor changes for sample code.
15. Minor bug fix.

Revision 3.00.000 (Released 2020-08-18)

1. Preliminary release version.

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